Khaled Hamdy Eldaoudy

Mechatronics Engineering Student | Embedded Systems & Robotics

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PROFESSIONAL SUMMARY

Detail-oriented Mechatronics Engineering student (Year 4) with hands-on experience in embedded systems, control, and robotics. Led 10+ technical projects in competitive settings combining C/C++, Python, and CAD. Completed 400+ hours of embedded systems, machine learning, and industrial training, including intensive programs at the Information Technology Institute (ITI) and the National Telecommunication Institute (NTI). Entrepreneurial background in e-commerce, digital marketing, and graphic design. Seeking an internship to apply multidisciplinary skills and a rapid-learning mindset to real-world engineering challenges.

EDUCATION

Bachelor of Mechatronics Engineering (Credit-Hour System) *2022 – Present*

Zagazig National University (ZNU), Egypt | Currently in 4th Year | Expected Graduation: 2026

TECHNICAL SKILLS

Embedded Systems & Firmware: Arduino, ESP32, STM32, AVR, Microcontrollers, C/C++, Embedded C

Control & Simulation: PID, MATLAB, Simulink

Robotics: Mechanical design, sensor fusion, sensor integration, autonomous navigation, motion control, robot design

Programming & Data: Python, Pandas, NumPy, Matplotlib, basic Machine Learning

CAD & 3D Design: SolidWorks (3D modeling, assemblies, simulation)

Productivity Tools: Microsoft Word, Excel, PowerPoint, Outlook

INDUSTRIAL & TECHNICAL TRAINING

Industrial Visit – Labtronic Industries *Apr – May 2026*

6th of October City, Egypt | Aerospace training equipment & engineering educational systems

- Represented the Engineering Student Union on an industrial visit exploring real-world aerospace and mechanical engineering applications
- Toured specialized labs including a Wind Tunnel, Cabin Simulator, Landing Gear Mechanism, and Aero Components Stability and force-measurement systems (velocity, lift, drag, thrust)
- Gained hands-on exposure to engine operation/testing systems and various control, operation, and lighting systems
- Helped coordinate the visit to strengthen ties between academic learning and industrial practice for engineering students

Information Technology Institute (ITI), Smart Village, Egypt *Jul – Aug 2025*

Embedded Systems Summer Training Program (162 Hours)

- Completed an intensive, hands-on training program in Embedded Systems Design and Development
- Covered C Programming, Microcontroller Interfacing, Communication Protocols, Real-Time Operating Systems (RTOS), and Embedded Systems Testing
- Designed, built, and defended a team capstone project — a Smart Home Control System — demonstrating system integration and real-time control (see Personal & Academic Projects)
- Supervised by Dr. Heba Saleh, Chairman of ITI

National Telecommunication Institute (NTI), in cooperation with ITIDA *Jul – Aug 2025*

Machine Learning Summer Training Program (120 Hours: 90 Technical + 30 Freelancing)

- Completed an intensive summer training program covering Machine Learning fundamentals and applications
- Combined technical ML training with freelancing and professional skills development
- Achieved a final score of 89%

Obour Land for Food Industries – Training Center *Aug – Sep 2025*

Beginner Technical Training, Tetra Pak A3-Speed Machines (120 Hours)

- Gained practical experience in food packaging technology and aseptic processing
- Covered machine overview, utilities (power, steam, air, H2O2), and automatic splicing & tube forming
- Learned filling & sterilization systems, drive & jaw mechanisms, and final folder operations
- Participated in an on-site factory tour observing large-scale automation and production workflows

LEADERSHIP EXPERIENCE

President, Engineering College Student Union *2023 – Present*

Zagazig National University

- Represented the university and Faculty of Engineering at the 9th UniEgypt Higher Education Expo, showcasing Mechatronics and Civil Engineering projects; received commendation from the Minister for innovation
- Coordinate communication between students and faculty; organize student activities and events

Member, ZNU Robotics Community *2024 – Present*

Zagazig National University

- Co-founded the community and helped organize workshops, hackathons, competitions, and peer mentoring

Supervisor, "We Help You Build Yourself" Student Initiative *2026*

Engineering Student Union, Zagazig National University

- Supervised and coordinated the initiative's activities as a representative of the Engineering Student Union, ensuring smooth execution and a productive, motivating environment for participants
- Recognized with a Certificate of Appreciation for leadership, responsibility, and continuous support throughout the initiative

Representative, 2nd Leaders Forum – Canal & Sinai Universities Alliance *Feb 2026*

"Leaders by Will" | New Ismailia National University, in cooperation with Suez Canal University

- Represented Zagazig National University and the Engineering Student Union at a regional leadership forum bringing together student leaders from the Canal and Sinai universities alliance

SELECTED PROJECTS & COMPETITIONS

Error Competition 2025 – Sumo Robot (2nd Edition) *October 2025*

IEEE HTI Student Branch, Higher Technological Institute

- Competed as part of Team ZNUS under the ZNU Robotics Community, representing Zagazig National University
- Designed and programmed three autonomous sumo robots for Egypt's national Sumorobot qualifying competition for the international championship in Japan
- NASA Team, from the same community, achieved 2nd place among 20+ university teams
- Awarded a Certificate of Appreciation for technical contribution in robot design, programming, and control systems

RoboSoccer Competition *August 2025*
IEEE HTI Student Branch

- Secured 2nd place with Team ZNUS and 5th place with Team ZUNBots among top Egyptian universities
- Led control system design and tuning for stable, precise robot performance during matches
- Received a certificate signed by Egypt's Prime Minister, Mostafa Madbouly, for national-level recognition

National Robotics Competition *March 2025*
Ministry of Youth and Sports

- Optimized Embedded C++ code for motion control; honored for innovation and problem-solving by ministry officials

IEEE Egypt Section Team Recognition *March 2025*

- Represented the ZNU Robotics Team; recognized for strong team performance

Robot Sumo 3 *February 2025*
Egyptian Russian University

- Integrated AI decision logic on STM32 as a member of NASA Team
- Received Certificate of Appreciation for hardware/software integration

Robot Sumo Competition 4 *2025*
Egyptian Russian University – Huawei ICT Academy Branch

- Contributed to the design and development of an autonomous sumo robot for the competition's 4th edition
- Received a Certificate of Appreciation for contribution to the competition

Interactive Robotics Demo *2025*
Zagazig National University

- Co-organized the university's first live robotics showcase
- Delivered workshops on C++/Python programming and CAD

Scientific Participation – National Conference on Sustainable Development *May 2025*

- Presented sustainability ideas aligned with Egypt Vision 2030
- Earned a conference shield and participation certificate

Autonomous Robot, IEEE Robotics Competition *2023*

- Designed and built a line-following robot with obstacle avoidance
- Implemented PID control; ranked 4th of 30+ teams

PERSONAL & ACADEMIC PROJECTS

Smart Home Control System – ITI Capstone Project (Team of 3) *2025*
ATmega32 Microcontroller (AVR) | C (AVR-GCC), Proteus

- Designed a smart home system to control home appliances automatically via sensors or manually via Bluetooth
- Implemented a keypad-based password/security system with LCD display feedback, and Bluetooth control (HC-05 module via UART) from a mobile terminal app
- Integrated a temperature sensor (LM35) and an LDR via ADC, a relay module to switch appliances, and PWM-driven motor/LED control
- Built and validated the full circuit in Proteus before hardware implementation; presented and defended the project to ITI instructors

Robotic Arm – Pick-and-Place System

- Designed and built a multi-degree-of-freedom robotic arm using Arduino and servo motors for automated pick-and-place operations
- Programmed precise motion control and object handling, applying concepts of robotics, kinematics, and embedded systems
- Built for educational and industrial simulation purposes, integrating mechanical assembly, electronics, and software control

Earthquake Simulation House

- Designed an Arduino UNO-based earthquake simulator to demonstrate the effects of seismic vibration on building structures
- Controlled vibration intensity via a potentiometer-driven PWM signal to an L298N motor driver and GA25-370 DC gear motor, with a custom mechanical cam mechanism generating variable shaking levels
- Integrated an SW-420 vibration sensor, LED indicators, and an SG90 servo motor to simulate automatic emergency response
- Built as an educational demonstration tool for civil engineering, mechatronics, and disaster-awareness applications

Vibration Alarm & Smart Safety System

- Built a smart vibration-based security and alarm system using Arduino UNO and an SW-420 sensor to detect abnormal movement or intrusion attempts
- Programmed real-time monitoring logic to trigger buzzers, warning LEDs, and an SG90 servo-actuated safety door mechanism upon detecting excessive vibration
- Demonstrates integration of embedded systems, sensors, actuators, and automated emergency response for smart security applications

ENTREPRENEURIAL & FREELANCE EXPERIENCE

Founder & Manager, CTRL Online Store *2024 – Present*

E-commerce brand selling electronics and accessories (Wuilt platform)

- Manage product listings, SEO optimization, and content creation
- Lead social media marketing and advertising strategy across TikTok, Facebook, and Instagram

Founder & E-Commerce Manager, Electronic Accessory Stores *2019 – 2022*

- Launched and ran 3 Shopify/WooCommerce stores; managed sourcing, fulfillment, and customer support
- Generated consistent sales across multiple product cycles

Digital Marketer & Media Buyer (Freelance) *2019 – 2022*

- Created and optimized paid ads on Facebook and Instagram
- Improved ROAS through targeting, A/B testing, and funnel analysis

Graphic Designer (Freelance) *2019 – 2022*

- Designed social-media visuals and ad creatives using Photoshop and Canva
- Maintained brand consistency for small businesses

Product Seller & Affiliate Marketer, Noon & Amazon Egypt *2019 – 2022*

- Listed trending tech/accessory products; applied SEO and FBA workflows
- Analyzed customer behavior and optimized product listings

CERTIFICATIONS

- SolidWorks Workshop — 2B Digital (March 2026)
- Machine Learning Summer Training (120 Hours) — NTI / ITIDA (Jul–Aug 2025)
- Certificate of Achievement, End-of-Term-1 Student Activities — Zagazig National University (2026)
- Error Competition 2025 (Sumo Robot) — IEEE HTI Student Branch (Oct 2025)
- C Programming: From Basics to Mastery — MaharaTech (Aug 2025)
- Artificial Intelligence — The British University in Egypt (2023)
- Introduction to AI & Applications — Zewail City of Science and Technology (2023)
- 3DExperience SOLIDWORKS Essentials Seminar — Dassault Systèmes / EUE (2025)
- SolidWorks Associate — Udemy (2022)
- Programming & Data Analysis — Coursera and Satr (2022 – 2024)

LANGUAGES

Arabic — Native English — B2 (strong reading and writing; conversational speaking)

SOFT SKILLS

Teamwork | Problem Solving | Time Management | Leadership | Communication